

Maintenance Manual — Helix Pharmaceuticals Building 4

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Plant	Helix Pharmaceuticals Building 4	Effective Date	15 Jan 2026
Document Owner	D. Petrov, Facilities & Engineering Lead	Classification	Controlled — Site Use

1. Purpose & Scope

This manual defines preventive, calibration, and qualification maintenance for cleanroom HVAC, isolator barriers, autoclaves, lyophilizers, and water systems supporting GMP production at Helix Pharmaceuticals Building 4. All activities are GxP-relevant and managed under the eQMS.

2. Critical Asset Register

The table below lists the critical assets covered by this manual. Each asset has a detailed task package in the CMMS (SAP-PM / Maximo) linked to its asset ID.

Asset ID	Description	Maintenance Strategy
HVAC-B4-01	Grade B AHU with HEPA terminal	DOP/PAO test annually; airflow visualization every 2 years.
ISO-04	Filling line isolator	Glove integrity test per batch; H2O2 vaporization cycle qualified annually.
WFI-01	Water-for-Injection still & loop	Continuous TOC and conductivity; sanitization at 85 °C weekly.
AUT-12	Pure-steam autoclave	Bowie-Dick test daily; IQ/OQ requalification every 12 months.
LYO-02	Production lyophilizer	Vacuum integrity test before each campaign; shelf temperature mapping annually.

3. PM Schedule Summary

Frequency	Activity
Per batch	Glove integrity, leak rate on isolator ISO-04.
Daily	Bowie-Dick on AUT-12; WFI sample point rotation.
Weekly	WFI loop sanitization; non-viable particle review for Grade B.
Quarterly	Calibration of critical pressure and temperature transmitters.
Annual	DOP/PAO testing; isolator H2O2 cycle re-qualification; lyophilizer mapping.

4. Work Management

All maintenance work is authorized through a CMMS work order. Work orders are categorized as PM, PdM, Corrective, Emergency, or Project. Maintenance KPIs (Schedule Compliance ≥85%, PM Compliance ≥95%, MTBF) are reported weekly to the Plant Manager.

5. Procedures

5.1 Change Control

Any maintenance that alters equipment or system performance follows the Change Control SOP CC-HLX-01. Like-for-like part swaps documented in the CMMS without formal change control.

5.2 Calibration

Calibration intervals defined per criticality (A = quarterly, B = annual, C = 2-year). Out-of-tolerance events trigger an impact assessment under deviation DEV-HLX-100.

5.3 Computerized Systems

Any maintenance affecting GxP computerized systems triggers a 21 CFR Part 11 impact assessment. Audit trail review is mandatory after firmware updates.

6. References

- ISPE Baseline Guide Vol. 3 — Sterile Product Manufacturing
- USP <1116> — Microbial Control
- FDA Guidance — Sterile Drug Products by Aseptic Processing (2004)
- Internal: CC-HLX-01, DEV-HLX-100, CAL-HLX-MS-04

7. Revision History

Rev	Date	Author	Summary
1	04 Jun 2023	D. Petrov	Initial issue.
2	21 Nov 2024	D. Petrov	Added predictive analytics integration.
3	09 Jul 2025	D. Petrov	Updated asset register.
4	15 Jan 2026	D. Petrov	Refreshed PM frequencies and KPIs.

Approvals

Role	Name	Signature	Date
Maintenance Engineering	D. Petrov, Facilities & Engineering Lead	_____	15 Jan 2026
Reliability Engineering	Sr. Reliability Engineer	_____	15 Jan 2026
Operations / Site Lead	A. Nakamura, Site Head	_____	15 Jan 2026